

$$\begin{aligned}
& \min_{P,E,Z,Q} \sum_{s=1}^S \|Z_s\|_* + \sum_{s=1}^S \lambda \|E_s\|_{2,1} + \sum_{s=1}^S \gamma \|P_s X Q_s - P_s X Q_s Z_s\|_F^2 + \beta \sum_{s \neq t} \text{HSIC}(Z_s, Z_t) \\
& \text{s.t. } X Q_s = X Q_s Z_s + E_s, P_s^T P_s = I \\
& \min_{P,Z,Q} \sum_{s=1}^S \|Q_s\|_F^2 + \sum_{s=1}^S \|X_s - X Q_s\|_F^2 + \sum_{s=1}^S \gamma \|P_s X_s - P_s X_s Z_s\|_F^2 \\
& \text{s.t. } X_s = X_s Z_s + E_s, P_s^T P_s = I
\end{aligned}$$

其中：

$X = [X_1, X_2, \dots, X_S] \in \mathbb{R}^{D \times N}$, 是多模态数据, D 为原始数据的特征数, N 为所有模态数据的样本总数 $N = \sum^S n_s$

$X_s \in \mathbb{R}^{D \times n_s}$, 是每个模态的数据

$P_s \in \mathbb{R}^{d \times D}$, d 为降维之后特征数

$Q_s \in \mathbb{R}^{N \times n_s}$, n_s 为单个模态数据的样本数

$Z_s \in \mathbb{R}^{n_s \times n_s}$

$E_s \in \mathbb{R}^{D \times n_s}$

↓ 堆叠

$$\begin{aligned}
& \min_{P,E,Z,Q} \|Z\|_* + \lambda \|E\|_{2,1} + \gamma \|P X Q - P X Q Z\|_F^2 + \beta \sum_{s \neq t} \text{HSIC}(Z_s, Z_t) \\
& \text{s.t. } X Q = X Q Z + E, P^T P = I
\end{aligned}$$

$X \in \mathbb{R}^{D \times N}$, D 为原始数据的特征数, N 为所有模态数据的样本总数 $N = \sum^S n_s$

$P = [P_1; P_2; \dots; P_S] \in \mathbb{R}^{(S \times d) \times D}$

$Q = [Q_1, Q_2, \dots, Q_S] \in \mathbb{R}^{N \times \sum n_s}$, n_s 为单个模态数据的样本数

$Z = \text{diag}[Z_1, Z_2, \dots, Z_S] \in \mathbb{R}^{\sum n_s \times \sum n_s}$

$E = [E_1, E_2, \dots, E_S] \in \mathbb{R}^{D \times \sum n_s}$

↓ 优化

$$\begin{aligned}
& \min_{P,E,Z,Q,J} \|J\|_* + \lambda \|E\|_{2,1} + \gamma \|P X Q - P X Q Z\|_F^2 + \beta \sum_{s \neq t} \text{HSIC}(Z_s, Z_t) \\
& \text{s.t. } X Q = X Q Z + E, Z = J, P^T P = I
\end{aligned}$$

增广拉格朗日函数

$$\begin{aligned}
\mathcal{L}(P, E, Z, Q, J, \Lambda, \Pi) = & \|J\|_* + \lambda \|E\|_{2,1} + \gamma \|P X Q - P X Q Z\|_F^2 + \beta \sum_{s \neq t} \text{HSIC}(Z_s, Z_t) \\
& + \langle \Lambda, X Q - X Q Z - E \rangle + \frac{\mu}{2} \|X Q - X Q Z - E\|_F^2 \\
& + \langle \Pi, Z - J \rangle + \frac{\mu}{2} \|Z - J\|_F^2
\end{aligned}$$

↓ 化简

$$\begin{aligned}\mathcal{L}(P, E, Z, Q, J, \Lambda, \Pi) = & \|J\|_* + \lambda\|E\|_{2,1} + \gamma\|PXQ - PXQZ\|_F^2 + \beta \sum_{s \neq t} \text{HSIC}(Z_s, Z_t) \\ & + \langle \Lambda, XQ - XQZ - E \rangle + \langle \Pi, Z - J \rangle \\ & + \frac{\mu}{2} \left(\|XQ - XQZ - E\|_F^2 + \|Z - J\|_F^2 \right)\end{aligned}$$

↓ 化简

$$\begin{aligned}\mathcal{L}(P, E, Z, Q, J, \Lambda, \Pi) = & \|J\|_* + \lambda\|E\|_{2,1} + \gamma\|PXQ - PXQZ\|_F^2 + \beta \sum_{s \neq t} \text{HSIC}(Z_s, Z_t) \\ & + \frac{\mu}{2} \left(\left\| XQ - XQZ - E + \frac{\Lambda}{\mu} \right\|_F^2 + \left\| Z - J + \frac{\Pi}{\mu} \right\|_F^2 \right)\end{aligned}$$

$X \in \mathbb{R}^{D \times N}$, D为原始数据的特征数, N为所有模态数据的样本总数 $N = \sum^S n_s$

$P = [P_1; P_2; \dots; P_s] \in \mathbb{R}^{(S \times d) \times D}$

$Q = [Q_1, Q_2, \dots, Q_s] \in \mathbb{R}^{N \times \sum n_s}$, n_s 为单个模态数据的样本数

$Z = \text{diag}[Z_1, Z_2, \dots, Z_s] \in \mathbb{R}^{\sum n_s \times \sum n_s}$

$E = [E_1, E_2, \dots, E_s] \in \mathbb{R}^{D \times \sum n_s}$

$$\begin{aligned}\mathcal{L}(P_s, E_s, Z_s, Q_s, J_s, \Lambda_s, \Pi_s) = & \sum_{s=1}^S \|J_s\|_* + \sum_{s=1}^S \lambda\|E_s\|_{2,1} + \sum_{s=1}^S \gamma\|P_s X Q_s - P_s X Q_s Z_s\|_F^2 + \sum_{s=1, s \neq t}^S \beta \text{HSIC}(Z_s, Z_t) \\ & + \sum_{s=1}^S \frac{\mu}{2} \left(\left\| X Q_s - X Q_s Z_s - E_s + \frac{\Lambda_s}{\mu} \right\|_F^2 + \left\| Z_s - J_s + \frac{\Pi_s}{\mu} \right\|_F^2 \right)\end{aligned}$$

2. Z

3. P

$$\min_P \gamma \|PXQ - PXQZ\|_F^2$$

4. Q

$$\min_Q \gamma \|PXQ - PXQZ\|_F^2 + \frac{\mu}{2} \left\| XQ - XQZ - E + \frac{\Lambda}{\mu} \right\|_F^2$$